17 November 1976

MILITARY SPECIFICATION

CLOTH, TWILL, COTTON, 10.0 OUNCES (339 g)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for cotton twill cloth.

1.2 Classification. The cloth shall be of the following classes, as specified (see 6.2):

Class 1 - Natural or bleached
Class 2 - Natural or bleached, mildew-resistant treated

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. The cloth shall match the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton which has been carded, drawn and spun into single yarn.

3.3 Color. The color of the cloth shall be natural or bleached. The color of the class 2 cloth shall be that shade imparted to the cloth as a result of the mildew-resistant treatment.

3.4 Physical requirements. The treated cloth shall conform to the physical requirements shown in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Weight oz./sq. yd. min. (g/m^2)	Yarns per inch (cm) (min.) Warp	Filling	Breaking strength (min.) pounds (newton) Warp	Filling
10.0 (339)	85 (33)	38 (15)	145 (645)	125 (556)

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3.4.1 Weave. The weave shall be 3 up and 1 down twill.

3.4.2 Width. The width shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.5 Finish. The cloth shall be finished so that it will not unravel when cut.

3.6 Nonfibrous materials. The cloth shall not exceed 15.0 percent starch and protein content including chloroform-soluble and water-soluble material when tested as specified in 4.4.

3.7 Mildew resistance (class 2). The class 2 cloth shall be mildew-resistant treated with copper-8-quinolinolate so as to deposit a copper content of 0.15 (± 0.05) percent metallic copper from copper-8-quinolinolate when tested as specified in 4.4.

3.8 Length and put up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (37 m). Each length shall be put up on full width rolls as specified in PPP-P-1134.

3.9 Fiber identification. Each roll of cloth shall be labeled and ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.10 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

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4.2.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.2.

4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined, and defects clearly noticeable at normal inspection distance (3 feet) (.91 m) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except when two or more defects represent a single local condition of the cloth, only one shall be scored. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be one linear yard (.91 m). The acceptance quality level shall be 6.0 defects per 100 units (yards) (meters). The sample size shall be in accordance with level I of MIL-STD-105. The lot size shall be expressed in units of one yard (.91 m) each. The number of rolls from which the sample is to be selected shall be in accordance with table II. An approximately equal number of yards shall be examined in each roll in the sample.

Defects
Hole, cut or tear
Smash
Knot or slub 1/
Spot or stain
Embedded crease or wrinkle
Two or more yarns missing, or broken for more than nine inches (22.9 cm)

1/ To be scored when larger than limits shown on Sears Fabrics Defect Scales (see 6.4) "T" for slubs and "D" for knots.

4.2.2.2 Overall examination. Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in table II.

Defects
Width less than specified 1/
Overall uncleanness

1/ Shall not be scored unless it occurs for more than 5 yards (4.6 m) per roll.

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TABLE II. Sample size

Lot size (yards) (.91 meter)	Sample size (rolls)	Acceptance number
Up to and including 3200 (2,926 m)	3	0
3201 (2,927 m) up to and including 10,000 (9,144 m)	5	0
10,001 (9,145 m) up to and including 35,000 (32,004 m)	8	1
35,001 (32,005 m) up to and including 150,000 (137,160 m)	13	2
150,001 (137,161 m) and over	20	3

4.2.2.3 Examination for length.

4.2.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards (1.8 m) less than the length marked on the ticket, shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.2.2.4 Examination for compliance with Textile Fiber Products Identification Act. An examination shall be made to insure that each roll is labeled for fiber content in accordance with the Textile Fiber Products Identification Act.

4.3 Packaging inspection. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with the section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3, except where otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized

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in expressing the final result. The sample unit shall be 2 yards (1.8 m), full width of the cloth. The lot size shall be expressed in units of one yard (.91 m). The sample size shall be as follows and the lot size shall be unacceptable if one or more sample units fail to meet any requirements specified.

Lot size (yards) (.91 meters)	Sample size (units)
800 (731 m) or less	2
801 (732 m) up to and including 22,000 (20,117 m)	3
22,001 (20,118 m) and over	5

TABLE III. Test methods

Characteristic	Requirement paragraph	Test method
Yarn:		
Cotton identification	3.2.1	1200 1/
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4.1	Visual 1/
Nonfibrous materials	3.6	2611
Copper content (class 2)	3.7	2050
Copper-8-quinolinolate (class 2)	3.7	2/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The contractor shall certify that only copper-8-quinolinolate was used in the treatment of the cloth.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and preserved-packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C packing. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth is intended for use in the lining for boots and shoes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class of cloth required (see 1.2).
(c) Width of cloth required (see 3.4.2).
(d) Length of roll required if other than specified (see 3.7).
(e) Selection of applicable put-up, preservation-packaging, and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample address the procuring office issuing the invitation for bids.

6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 733Q, "Sears Defect Replica Kit", Sears Tower, Chicago, IL 60684.

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activity:

Air Force - 99

Preparing activity:

Army - GL

Project No. 8305-0555

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-43956

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MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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